

Multigene sequence data reveal morphologically cryptic phylogenetic species within the genus *Laccaria* in southern Australia

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Abstract: *Laccaria* (Hydnangiaceae, Agaricales, Basidiomycota) is one of the more intensively studied ectomycorrhizal genera; however, species boundaries within *Laccaria* and the closely related *Hydnangium* and *Podohydangium* in Australia have not yet been examined with molecular sequence data. Based on morphological characters, eight native species of *Laccaria* are currently recognized in Australia, as well as three *Hydnangium* species and the monotypic *Podohydangium australe*. Sequences of the internal transcribed spacer region of nuclear rDNA (ITS), RNA polymerase beta subunit II (*rpb2*) and translation elongation factor 1 alpha (*tef1α*) were generated from 77 collections of *Laccaria*, *Hydnangium* and *Podohydangium* from Australia. Ten phylogenetic species and a further 11 potential species (represented by singletons) of *Laccaria* in Australia are delimited from sequence analyses. Most of the morphological species contained cryptic phylogenetic species, but these species were always nested entirely within a given morphological species, although not always as sister taxa. The *rpb2* locus performed best as a species barcode with pairwise and patristic distance measures. The ITS sequence region returned the least resolved gene tree of the three regions examined and was the least useful as a barcode region. Based on the phylogenetic topology, there appears to have been multiple gains and/or losses of the ectomycorrhizal association of some species with the myrtle beech, *Nothofagus cunninghamii* as well as of sequestrate basidiocarps and two-spored basidia.

Key words: *Hydnangium*, *Podohydangium*, *rpb2*, sequestrate fungi, *tef1α*

INTRODUCTION

Laccaria species have been used to study many aspects of ectomycorrhizal biology, owing in part to their ability to grow in culture as well as their ubiquity in nature. In Australia, native *Laccaria* species are widespread and typically grow in forests dominated by trees such as *Eucalyptus*, Casuarinaceae, *Leptospermum* and *Acacia*, although some species have been recorded only in cool, temperate rainforests where *Nothofagus cunninghamii* is the dominant potential host.

Laccaria belongs to Hydnangiaceae within the agaricoid clade of the Agaricales (Matheny et al. 2006). While morphological similarities between the agaricoid (pileate-stipitate) *Laccaria*, the sequestrate (truffle-like) *Hydnangium* (and later the intermediary *Podohydangium*) have been recognized for more than 30 y, the phylogenetic relationships between the three have not always been agreed upon (Kühner 1980, Beaton et al. 1984, Singer 1986, Mueller et al. 1993). Mueller and Pine (1994) were the first to use molecular sequence data (from the second internal transcribed spacer of rDNA, ITS2) to support the monophyly of a combined *Laccaria*, *Hydnangium* and *Podohydangium* group, but the genera were not formally synonymized pending data from further genes.

While *Laccaria* is one of the most readily identifiable genera within the Agaricales (Mueller 1992), delimitation of species on morphological characters has proved challenging. May (1991) examined 20 macromorphological characters and 12 micromorphological characters in multivariate analyses (phenetic classification and ordination) of 64 collections of native *Laccaria* from predominately southern and eastern Australia. As a result, eight native species (*L. lateritia*, *L. canaliculata*, *L. masoniae*, *L. spp. A, B, C, D* and *E*) were circumscribed. While each of these species had a characteristic set of macro- and micromorphological characters, many pairs of species overlapped on most characters and some could be distinguished only by the multivariate analyses. Some pairs of species differed primarily by production of two- or four-spored basidia (e.g. *L. lateritia*, *L. canaliculata*; May 1991). This character has been used historically to separate species within *Laccaria*, and the practice has been maintained more recently (Osmundson et al. 2005).

Where species delimitation is very difficult based on morphological characters, molecular sequence data provide characters that may be more discriminating. Sequence data also may be useful as DNA barcodes for the species they distinguish, as well as reveal cryptic variation in species originally thought to be morphologically uniform (Hawksworth 2006), which can lead to the recognition of new species (e.g. *Pisolithus*, Martin et al. 2002; *Lactifluus*, Van de Putte et al. 2012). Including information from multiple gene regions is recommended in that individual genes may be subject to processes such as incomplete lineage sorting, horizontal gene transfer or gene duplication and extinction (Maddison 1997). To this end, the genealogical concordance phylogenetic species recognition (GCPSR) method (Taylor et al. 2000) aims to represent more closely the underlying species tree by incorporating multiple gene trees rather than single gene trees.

The aim of this study was to delimit species of *Laccaria* and the closely related *Hydnangium* and *Podohydangium* from southern Australia with a multigene (ITS, *rpb2*, *tef1 α*) phylogeny identified against the existing morphological classification. These three gene regions then were assessed for their utility as DNA barcoding regions. Selected ITS sequences from GenBank (from Australia, New Zealand or associated with exotic *Eucalyptus*) were analyzed to detect further cryptic species. The implications of the phylogeny on the evolution of host preference, sequestration and two-spored basidia were considered, using the phylogenetic species delimited in the multigene study.

MATERIALS AND METHODS

Field sampling and herbarium specimens.—Fresh basidiocarps of Australian *Laccaria* and *Hydnangium* associated with native tree hosts were collected from 2009 to 2011, resulting in 57 collections (TABLE I). Basidiocarps were dried with warm air using a food dehydrator, then vouchered in the National Herbarium of Victoria (MEL). A further 17 dried specimens and three cultures representing native as well as exotic species (*L. proxima*, *L. bicolor*) were analyzed (TABLE I).

Microscopy.—Dried tissue from voucher specimens was rehydrated in 3% KOH, mounted in Congo red (in a dilute alkali solution) and observed under the light microscope. Collections (TABLE I) were assigned a species designation by identification against the macro- and micromorphological characters of the seven native species described or reported from Australia (Beaton et al. 1984, Bougher et al. 1993) and the five undescribed species distinguished by May (1991, 1997; SUPPLEMENTARY TABLE I). Some collections had one or two morphological character states that were outside the range typical for the species to which they would otherwise have been assigned; those collections were indicated by “cf.”.

Genomic DNA isolation.—A piece of stipe, pileus or gleba (2–20 mg) was taken from herbarium material, silica-gel dried fresh collections or mycelium from liquid cultures. Tissue was freeze-dried, and the DNA extracted by tissue maceration, incubation in 2% cetyltrimethylammonium bromide (CTAB) buffer for 1 h at 65 C. Protein was removed with phenol:chloroform (1:1) and chloroform, DNA precipitated with isopropanol and resuspended in 50 μ L of TE buffer (pH 8) containing RNase. DNA was diluted 1:10–1:200 as required for use.

PCR amplification and sequencing.—The internal transcribed spacer (ITS) of nuclear ribosomal DNA, the region between conserved domains 6 and 7 of RNA polymerase II subunit 2 (*rpb2*) and the 3' end of translation elongation factor 1- α (*tef1 α*) were amplified with primers (TABLE II). The PCR conditions used to amplify ITS and *rpb2* were denaturation at 94 C for 3 min; 34 cycles at 94 C for 30 s, annealing temperature for 30 s, 72 C for 2 min and a final elongation at 72 C for 10 min. Annealing temperatures were 50 C for ITS and 58 C for *rpb2*. The touchdown PCR conditions used to amplify the fragment of *tef1 α* were: 94 C for 3 min; 40 cycles at 94 C for 30 s, 65 C for 30 s (decreasing by 1 C every cycle for the first 15 cycles) and 72 C for 1 min; and a final elongation at 72 C for 10 min. Amplification products were sequenced at the Australian Genome Research Facility.

Bioinformatic analyses.—Sequence trace files were assembled and analyzed with Sequencher 4.9 (Gene Codes Corp., Ann Arbor, Michigan). Sequences were aligned in MEGA 4 (Tamura et al. 2007) with Clustal W (Thompson et al. 1994), followed by manual adjustment and removal of ambiguously aligned regions.

Alignments were constructed for the three gene regions separately and combined. The combined alignment included Australian collections for which at least two of the three gene regions had been sequenced and eight northern hemisphere *Laccaria* collections for seven of which only the ITS was available. A second ITS alignment was constructed for an expanded dataset that included sequences for which only the ITS was available. Twenty of the additional ITS sequences were obtained from GenBank with searches using the terms *Laccaria*, *Hydnangium* and *Podohydangium* with Australia, New Zealand and *Eucalyptus*, which is a tree genus widely planted around the world but for which nearly all species are native to Australia. In the multigene, *rpb2* and *tef1 α* analyses the outgroup taxa were *Mythicomycetes corneipes* (Psathyrellaceae), *Agaricus bisporus*, *Lycoperdon pyriforme* and *Coprinus comatus* (all Agaricaceae). In both ITS analyses only *M. corneipes* was used as outgroup. Sequences generated in the present study were deposited in GenBank (TABLE I), and phylogenetic trees and datasets were deposited in TreeBASE (<http://purl.org/phylo/treebase/phyloids/study/TB2:S13533>).

Phylogenetic analyses.—Phylogenies were constructed for each gene alignment individually (SUPPLEMENTARY FIGS. 1, 2, 3). Tree topologies were assessed visually for congruence of species-rank clades (see below for species level classification), and because there were no strongly supported

TABLE I. Collections of fungi used in the phylogenetic analyses

Collection ID	Morphological species designation ^a	Phylogenetic species designation ^b	Location ^c	Herbarium no. (MEL) ^d	GenBank accession nos.		
					ITS	<i>rpb2</i>	<i>tef1α</i>
AC27829	<i>H. carneum</i>	HCARI	Chiltern Box-Ironbark NP, VIC	2271753	JX270694	JX295580	JX305593
GMM7271	<i>H. carneum</i>	HCARI	Kinglake, VIC	2293605	JX270755	JX295619	JX305630
RH8879	<i>H. carneum</i>	HCARI	Bega Valley, NSW	2297093	JX270758	JX295621	JX305632
TL2358	<i>H. carneum</i>	HCARI	Lake Elizabeth, VIC	2359675	JX270696	JX295622	
TL2359	<i>H. carneum</i>	HCAR2	Lake Elizabeth, VIC	2359676	JX270733		JX305633
TL2361	<i>H. carneum</i>	HCARI	Lake Elizabeth, VIC	2359677	JX270705	JX295623	JX305634
PBM3209	<i>H. carneum</i>	HCARI	Hobart, TAS		HQ832445		
K(M)164234	<i>H. carneum</i>	HCAR3	Italy		JN235948		
KS1494/06	<i>H. sublamellatum</i>	HSUB	Hopetoun, WA	2297141	JX270750	JX295620	JX305631
KHNZ06123	<i>Hydnangium</i> sp.	NZ2	New Zealand		FJ168595		
ES054	<i>L. canaliculata</i>	CAN1	Philosopher Falls, TAS	2359682	JX270722	JX295582	JX305595
ES070	<i>L. canaliculata</i>	CAN1	Philosopher Falls, TAS	2359683		JX295583	JX305596
ES281	<i>L. canaliculata</i>	CAN4	Lake St Clair, TAS	2359686	JX270723		JX305622
TM1811	<i>L. canaliculata</i>	CAN1	Blue Tier, TAS	2360196	JX270720	JX295625	JX305636
TM650	<i>L. canaliculata</i>	CAN1	Acheron Way, VIC	226545	JX270748		
ES093	<i>L. canaliculata</i>	CAN2	Wirrawilla, VIC	2359684	JX270714	JX295587	JX305600
ES187	<i>L. cf. canaliculata</i>	CAN3	Somers Park, VIC	2359685	JX270697	JX295600	JX305613
PDD:95783	<i>L. cf. canaliculata</i>	NZ3	New Zealand		HQ533019		
ES089	<i>L. glabripes</i>	LAC	Nilgiris, India		AJ699075		
ES100Y	<i>L. lateritia</i>	LAT1	Wirrawilla, VIC	2359687	JX270751	JX295585	JX305598
ES101Y	<i>L. lateritia</i>	LAT1	Fenshaw, VIC	2359688	JX270735	JX295589	JX305602
ES103O	<i>L. lateritia</i>	LAT1	Fenshaw, VIC	2359689	JX270719	JX295590	
ES103Y	<i>L. lateritia</i>	LAT1	Mait Rest, VIC	2359690	JX270688		JX305603
ES132	<i>L. lateritia</i>	LAT1	Fenshaw, VIC	2359691	JX270694		JX305604
GMM7243	<i>L. lateritia</i>	LAT1	Grampians NP, VIC	2359692	JX270736	JX295595	JX305609
ES172	<i>L. cf. lateritia</i>	LAT2	Australia		FJ168596		
BF2106	<i>L. masoniae</i>	MAS1	Black Mountain, ACT	2360194	JX270737	JX295599	JX305612
ES279	<i>L. cf. masoniae</i>	MAS2	Triplet Falls, VIC	2054578	JX270743		
ES285	<i>L. masoniae</i>	MAS1	Lake St Clair, TAS	2359693	JX270728	JX295610	JX305621
GM7199	<i>L. masoniae</i>	MAS1	Southwest NP, TAS	2359694	JX270741	JX295612	JX305624
TM1800	<i>L. masoniae</i>	MAS1	Somers Park, VIC	2293601		JX295618	JX305629
TM713	<i>L. masoniae</i>	MAS1	Blue Tier, TAS	2360195	JX270752	JX295624	JX305635
ES011	<i>L. sp. A</i>	SPA	Bulga NP, VIC	226406	JX270729		
ES014	<i>L. sp. A</i>	SPA	Philosopher Falls, TAS	2359645	JX270702		
			Philosopher Falls, TAS	2359646	JX270699		

TABLE I. Continued

Collection ID	Morphological species designation ^a	Phylogenetic species designation ^b	Location ^c	Herbarium no. (MEL) ^d	GenBank accession nos.		
					ITS	<i>rpb2</i>	<i>tef1α</i>
ES075	<i>L. sp. A</i>	SPA	Philosopher Falls, TAS	2359647	JX270721	JX295584	JX305597
ES076	<i>L. sp. A</i>	SPA	Philosopher Falls, TAS	2359648	JX270731		
ES083	<i>L. sp. A</i>	SPA	Philosopher Falls, TAS	2359657	JX270700		
ES087	<i>L. sp. A</i>	SPA	Philosopher Falls, TAS	2360498	JX270715		
ES193	<i>L. sp. A</i>	SPA	The Beeches, VIC	2359649	JX270738		
ES198	<i>L. sp. A</i>	SPA	The Beeches, VIC	2359650	JX270689	JX295602	
ES199	<i>L. sp. A</i>	SPA	The Beeches, VIC	2359651	JX270753		
ES210	<i>L. sp. A</i>	SPA	Tarra-Bulga NP, VIC	2359652	JX270692	JX295603	JX305615
ES232	<i>L. sp. A</i>	SPA	Aire River, VIC	2359653	JX270746	JX295604	JX305616
ES253	<i>L. sp. A</i>	SPA	Philosopher Falls, TAS	2359654	JX270691	JX295605	JX305617
ES261	<i>L. sp. A</i>	SPA	Savage River, TAS	2359656	JX270711	JX295606	
ES267	<i>L. sp. A</i>	SPA	Mt Pearce, TAS	2359655	JX270712	JX295607	JX305618
ES270	<i>L. sp. A</i>	SPA	Mt Read, TAS	2359660	JX270717	JX295608	JX305619
ES291	<i>L. sp. A</i>	SPA	Mt Field, TAS	2359661	JX270727	JX295613	JX305625
ES295	<i>L. sp. A</i>	SPA	Mt Michael, TAS	2359659	JX270693	JX295614	JX305626
ES312	<i>L. sp. A</i>	SPA	Weldborough Pass, TAS	2359658	JX270742	JX295615	JX305627
ES319	<i>L. sp. A</i>	SPA	Mt Horror, TAS	2359662	JX270703	JX295617	
ES322	<i>L. sp. A</i>	SPA	Ben Ridge Road, TAS	2359663	JX270687		
ES353	<i>L. sp. A</i>	SPA	Aire River, VIC	2359664	JX270734		
ES354	<i>L. sp. A</i>	SPA	Aire River, VIC	2359665	JX270716		
TM1014	<i>L. sp. A</i>	SPA	Acheron Way, VIC	226437	JX270686		
TM828	<i>L. sp. A</i>	SPA	Marysville, VIC	226430	JX270707		
ES105Y	<i>L. sp. B</i>	SPB1	Somers Park, VIC	2359666	JX270756	JX295591	JX305605
ES106	<i>L. sp. B</i>	SPB1	Dom Dom Saddle, VIC	2359667	JX270718	JX295592	JX305606
ES115	<i>L. sp. B</i>	SPB3	Grampians NP, VIC	2359668	JX270757	JX295593	JX305607
ES120	<i>L. sp. B</i>	SPB3	Grampians NP, VIC	2359669	JX270747	JX295594	JX305608
ES141	<i>L. sp. B</i>	SPB2	Tidbinbilla, ACT	2359670	JX270726	JX295596	JX305610
ES153	<i>L. sp. B</i>	SPB2	Tidbinbilla, ACT	2360193	JX270706	JX295597	
ES282	<i>L. sp. B</i>	SPB1	Lake St Clair, TAS	2359671	JX270744	JX295611	JX305623
ES318	<i>L. sp. B</i>	SPB1	Halls Falls, TAS	2359672	JX270759	JX295616	JX305628
TM1030	<i>L. sp. B</i>	SPB1	Bogong High Plains, VIC	226513	JX270708		
TM188	<i>L. sp. B</i>	SPB5	Tinderry Mountains, NSW	226440	JX270739		
TM819	<i>L. sp. B</i>	SPB1	Portland, VIC	226445	JX270690		
TM902	<i>L. sp. B</i>	SPB1	Otway Ranges, VIC	226451	JX270698		
TM927	<i>L. sp. B</i>	SPB1	Wilsons Promontory, VIC	226518	JX270730		
TM944 ^e	<i>L. sp. B</i>	SPB3	Mornington Peninsula, VIC	226520	JX270713	JX295628	JX305639

TABLE I. Continued

Collection ID	Morphological species designation ^a	Phylogenetic species designation ^b	Location ^c	Herbarium no. (MEL) ^d	GenBank accession nos.		
					ITS	<i>rpb2</i>	<i>tef1α</i>
	<i>L. montana</i>				EU486434		
	<i>L. proxima</i>				GQ267477		
Outgroups							
	<i>Mythomyces cornipes</i>					DQ408110	DQ029197
	<i>Lycoperdon pyriforme</i>					AY218495	AY883426
	<i>Coprinus comatus</i>					AY780934	AY881026
	<i>Agaricus bisporus</i>					AF107785	GU187673

^a Morphological species of the genera *Laccaria*, *Hydnangium* and *Podohydnum*; designations from observation of tissue (with MEL numbers) or from GenBank.
*Probably an Australian native that has been misidentified.
^b Phylogenetic species designations based on data in Figs. 1, 2.
^c Locations with state abbreviations are in Australia: NP = national park; VIC = Victoria; NSW = New South Wales; WA = Western Australia; TAS = Tasmania; ACT = Australian Capital Territory.
^d Gaps represent cases where sequences were derived from GenBank.
^e DNA extracted from cultures.
^f Culture of this collection was provided by INRA, Nancy, France.

TABLE II. Primers used to amplify the internal transcribed spacer (ITS) of nuclear rDNA, RNA polymerase beta subunit (*rpb2*) and translation elongation factor 1 alpha (*tef1* α)

Region	Primer name	Primer sequence 5'–3' ^a	Source
ITS	ITS1	TCCGTAGGTGAACCTGCGG	White et al. (1990)
	ITS4	TCCTCCGCTTATTGATATGC	White et al. (1990)
ITS1	ITS_18	TCGTAACAAGGTTTCCGTAGG	This study
	ITS_58_1	GGTGC GTTCAAAGATTCGAT	This study
ITS2	ITS_58_2	CAACGGATCTCTTGGCTCTC	This study
	ITS_28	TCCAGCGCGGATAATACTTC	This study
<i>rpb2</i> , intron 7	RPB2-6F_Lbic	TGGGGTATGGTGTGCCCTGC	Matheny (2005)
	RPB2-7.1R_Lbic	CCCATCGCTTGTTCCTCATAGC	Matheny (2005)
<i>rpb2</i> , introns 7, 8	fRPB2-5F_Lbic	GACGACCGTGATCATTTTGG	Matheny (2005)
	bRPB2-7R_Lbic	GACTGGTTATGATCGGGGAAAGG	Matheny (2005)
<i>tef1</i> α	EF1-983F_Lbic	GCCCCCTGGTCATCGTGATTTTCAT	Rehner and Buckley (2005)
	EF1-LspA-2R	GACGATGACCTGTGCGTTGA	This study

Primers designed for this study used sequences from *Laccaria bicolor* (DQ367906 and genome sequence, Martin et al. 2008), inhouse sequence of *L. sp. A* (ES295) and the software Primer3 (Rozen and Skaletsky 2000).

^aNucleotides that are underscored are modifications of the source primers and replace the original degenerate sites.

conflicts (collections were not placed with >75 bootstrap support in alternative terminal clades) among the three gene regions for each of the tree-building methods the alignments then were concatenated. Maximum likelihood (ML) analysis was conducted with RAxML HPC (Stamatakis et al. 2008) and used a general time reversible model with parameters for gamma and pInv (GTR+I+G). Each partition of the multigene alignment was modeled separately. Node support was estimated with 1000 bootstrap replicates. Bayesian inferences (BI) were performed with MrBayes 3.2 (Ronquist and Huelsenbeck 2003) specifying a GTR+I+G model and two separate runs of four chains (one cold, three heated) for 10 000 000 generations (sampling trees every 1000 generations) with a 25% burn-in. For the multigene analysis, each partition was modeled separately. Tracer 1.5 (<http://tree.bio.ed.ac.uk/software/tracer>) was used to confirm that split frequencies had converged between the two runs (<0.01) and the effective sample sizes were >200. The COMPARE and SPLITS functions (specifying 10 windows) in AWTY (Nylander et al. 2008) were used to confirm that tree topologies had also converged among and within the runs. Maximum parsimony (MP) analyses were performed in PAUP* (Swofford 1998) with gaps treated as a new character state and DELTRAN character-state optimization. Starting trees were by 1000 random addition replicates and subjected to TBR branch swapping with the MULTREES option on and a maximum of 1000 trees saved per replicate. Branch support was estimated by 1000 bootstrap replicates, each with 10 random addition replicates with a maximum of 500 trees saved for each addition replicate.

Morphological species are denoted by a capitalized abbreviation: *Hydnangium carneum* (HCAR); *H. sublamellatum* (HSUB); *Podohydangium australe* (PAUS); *Laccaria canaliculata* (CAN); *L. lateritia* (LAT); *L. masoniae* (MAS); *L. sp. A* (SPA); *L. sp. B* (SPB); *L. sp. C* (SPC); *L. sp. D* (SPD); and *L. sp. E* (SPE). Where molecular analysis recovered multiple phylogenetic species within a particular morphological species, each was designated by a number

after the capitalized abbreviation, as for example SPB1, SPB2, SPB3 etc.

Species boundaries.—Phylogenetic species boundaries were classified with a GCPSR approach (Taylor et al. 2000) where strongly supported nodes congruent across multiple gene regions were indicative of species and subdivision of such clades that would produce incongruent gene trees. Morphological species limits determined the level at which to expect species boundaries. The suitability of the three gene regions as species barcodes and the associated selection of cutoff values were assessed with comparisons of pairwise and patristic distance among and within species clades for the concatenated dataset as well as each of the three gene regions individually. Pairwise distance was calculated with MOTHUR (Schloss et al. 2009) with gaps treated as a single insertion. Pairwise distances were used to cluster sequences of ITS at 1, 2 and 3% dissimilarity. Patristic distance was calculated as the sum of the branch lengths between a pair of collections in the maximum likelihood trees, with the R package adephylo 1.1–2 (Jombart et al. 2010). The patristic distance measurement was included because it is less sensitive to character homoplasy/reversals between sequences and has been suggested as the preferred measure to use when phylogenetic information is available (Fourment and Gibbs 2006, Lefébure et al. 2006). The northern hemisphere and outgroup taxa were not included in calculations of distances.

RESULTS

Morphological data.—Among the 77 field and herbarium collections, all eight morphological species of Australian native *Laccaria* were represented, as well as two of the three *Hydnangium* species. Most collections fell within the range of variation established by May (1991), but three collections (ES172, ES279, ES187) closely resembled one of the morphological

species but had one or two characters that were outside the range typical for those species (SUPPLEMENTARY TABLE II).

Molecular data.—A total of 175 new sequences (75 ITS, 51 *rpb2*, 49 *tef1 α*) of Hydnangiaceae were produced from the 77 collections analyzed. The concatenated alignment contained 55 collections of Australian Hydnangiaceae for which sequences from at least two of the three gene regions were present, plus eight sequences of northern hemisphere species. The expanded ITS alignment included (in addition to the 53 ITS sequences in the concatenated alignment) a further 22 new and nine GenBank sequences from fungi collected in Australia, as well as three sequences that were from collections under *Eucalyptus* that were planted overseas, eight sequences from New Zealand Hydnangiaceae and eight sequences of northern hemisphere species.

The ITS alignment consisted of 591 base pairs after indels and ambiguously aligned regions (total of 72 bp) were excluded. The *rpb2* alignment was 590 bp with no introns. The *tef1 α* alignment contained three introns and was 752 bp after exclusion of an ambiguously aligned region in the first intron and the entire third intron, which was in the outgroup taxa only (total 95 bp).

Phylogenetic species recognition.—The three tree-building methods produced highly similar topologies for all three gene regions separately and combined. The terminal clades and the various supports they received in each of the trees are included (TABLE III). The major difference produced by the tree-building methods was in the placement of the collection ES281 (classified as *L. canaliculata*) with respect to the phylogenetic clade LAT1, which contained most of the *L. lateritia* sequences. Based on the *tef1 α* data, ES281 was placed either within (BI) or sister to LAT1 (ML, MP) and based on the multigene data ES281 was also either within (BI, ML) or sister to LAT1 (MP). The collection ES281 also differs from the remaining samples in the LAT1 clade on the basis of morphology. It produces four spores per basidium, while *L. lateritia* produces two. Thus it is considered here to be distinct from LAT1. The topology of the individual gene trees of *rpb2* and *tef1 α* (SUPPLEMENTARY FIG. 2, 3) did not vary substantially from that of the combined multigene phylogeny (FIG. 1), although the ITS gene tree differed in the breakdown of one of the clades of mixed morphological affinity (TABLE III).

Of the eight morphological species that were represented by more than one collection in the multigene phylogeny, only *L. sp. A* and *L. sp. D* formed monophyletic groups, although for the latter only three collections were analyzed (FIG. 1). Collec-

tions from all the remaining morphological species were present in multiple terminal clades. In most cases a core clade contained the majority of collections from each species with additional singletons that did not cluster together. However, for *L. sp. B* there were multiple collections in each of the three clades it formed, in addition to a singleton. There was no mixing of different morphological species in a given phylogenetic species clade. In no case did the morphological species that occurred in multiple genetic lineages form supported sister taxon relationships with their counterparts, despite the morphological similarity; instead clades were interspersed in the tree.

Four singleton collections from *L. sp. E* (ES154), *L. cf. canaliculata* (ES187), *H. carneum* (TL2359) and *L. cf. lateritia* (ES172) formed a supported clade labeled MIX1 (84% ML bootstrap, 1.00 posterior probability, 87% MP bootstrap). This mixed clade was not present in the ITS gene tree. Another clade that was highly supported in the multigene phylogeny and received no support in the ITS gene tree was labeled MIX2 (TABLE III). The MIX2 clade contained a supported two-collection subclade of *L. sp. B* (SPB2), another collection identified as *L. sp. B* (TM956) and a collection of *H. sublamellatum* (KS1494).

The higher level structure of the multigene phylogeny was of five well supported clades that would form a polytomy at the base of the tree if branches without strong support were collapsed. The first clade was the most diverse and contained the terminal clades LAT1, HCARI, SPD, CAN1, MIX1 and the *L. canaliculata* singleton (ES093). The second clade contained the sample of sequences from northern hemisphere species, the third contained SPB1 and *L. cf. masoniae* (ES279), the fourth contained SPA, SPE1 and MIX2 and the fifth contained MAS1 and SPB3. The northern hemisphere *Laccaria* species formed a monophyletic group either sister to (*rpb2*, *tef1 α*) or within (ITS) the Australian *Laccaria* (FIGS. 1, 2). *Laccaria sp. A* and the two phylogenetic species of *L. masoniae*, all of which are host specific to *Nothofagus*, did not fall within a single clade but occurred in three different places in the tree, in each case sister to species with generalist host affinities. The two-spored species *L. sp. E*, *L. lateritia*, *H. carneum* and *L. sp. C* also did not form a monophyletic clade, and no patterns were detected with respect to sister taxa.

Species boundaries based on multigene phylogeny.—Eighteen potential species were identified from the multigene phylogeny as either clades (FIG. 1) or singletons (SUPPLEMENTARY TABLE III). The median pairwise distances within each of the phylogenetic

TABLE III. Support (bootstrap and posterior probabilities) for each of the terminal clades as well as two mixed clades (MIX1 and MIX2) across all the gene partitions

Clade name	Support metric	Terminal clade probabilities				
		Gene regions				
		Multi-gene	<i>rpb2</i>	<i>tef1α</i>	ITS	ITS all
CAN1	ML	1.00	1.00	1.00	1.00	1.00
	PP	1.00	1.00	1.00	1.00	0.97
	MP	1.00	1.00	1.00	1.00	0.86
HCAR1	ML	1.00	1.00	1.00	0.93	0.95
	PP	1.00	1.00	1.00	1.00	1.00
	MP	1.00	0.98	1.00	0.80	0.67
LAT1	ML	1.00	0.98	0.57 ^a	0.93	0.89
	PP	1.00	1.00	1.00	1.00	0.98
	MP	0.90 ^a	0.90	0.92 ^a	0.97	0.93
MAS1	ML	1.00	1.00	1.00	0.72	0.52 para
	PP	1.00	1.00	1.00	0.90	0.93
	MP	1.00	1.00	1.00	poly	poly
SPA	ML	1.00	0.95	1.00	0.97	0.84
	PP	1.00	0.98	1.00	0.77	0.63
	MP	1.00	0.82	0.99	0.70	0.72
SPB1	ML	1.00	1.00	1.00	0.93	0.61
	PP	1.00	1.00	1.00	1.00	0.88
	MP	1.00	1.00	1.00	0.80	poly
SPB2	ML	1.00	1.00	na	0.99	0.99
	PP	1.00	1.00	na	1.00	1.00
	MP	1.00	0.96	na	0.99	0.98
SPB3	ML	1.00	1.00	1.00	0.95	0.82
	PP	1.00	1.00	1.00	0.99	1.00
	MP	1.00	1.00	1.00	0.93	0.82
SPD	ML	1.00	1.00	1.00	1.00	1.00
	PP	1.00	0.99	1.00	1.00	1.00
	MP	1.00	1.00	1.00	1.00	1.00
SPE1	ML	1.00	1.00	1.00	0.95	poly
	PP	1.00	1.00	1.00	0.80	0.81
	MP	1.00	1.00	1.00	0.91	0.91
MIX1	ML	0.84	0.82	0.75	poly	poly
	PP	1.00	0.95	0.99	poly	poly
	MP	0.87	0.69	0.81	poly	poly
MIX2	ML	0.99	0.95	0.69	0.48	0.55
	PP	1.00	0.99	0.97	0.90	0.93
	MP	1.00	0.98	0.64	poly	poly

rpb2: RNA polymerase beta subunit; *tef1 α* : translation elongation factor 1 alpha; ITS: internal transcribed spacer of nuclear rDNA; ITS all: expanded ITS dataset. ML = bootstrap support from 1000 replicates on a maximum likelihood tree, PP = posterior probability from 75 000 iterations of a Bayesian inference tree, MP = bootstrap support from 1000 replicates on a maximum parsimony tree. na = not applicable due to missing collections. para = paraphyletic; poly = polyphyletic. ^a = where LAT1 is monophyletic without collection ES281, the clade inclusive of ES281 and LAT1 scored 1.00 support.

species ranged from 0.1 to 0.7% dissimilarity (FIG. 3a). Data from the MIX1 and MIX2 clades, which both represent multiple species, were included and accordingly these clades had intraclade distances much greater than those for the accepted phylogenetic species (median values 2.4% and 1.5% respectively; FIG. 3). Patristic distances produced similar patterns of intraspecific variation to the pairwise distances, but the scale of the variation was generally

higher. The median patristic distances within each of the phylogenetic species was <0.05–1.3%. The mixed species clades MIX1 and MIX2 again had much greater median values of 5% and 4% respectively. When the *L. canaliculata* collection ES281 was included within LAT1, the upper limit of pairwise distances within the LAT1 clade increased 0.9–1.6%. Calculations of patristic distances are dependent on the tree topology, and collection ES281 was not

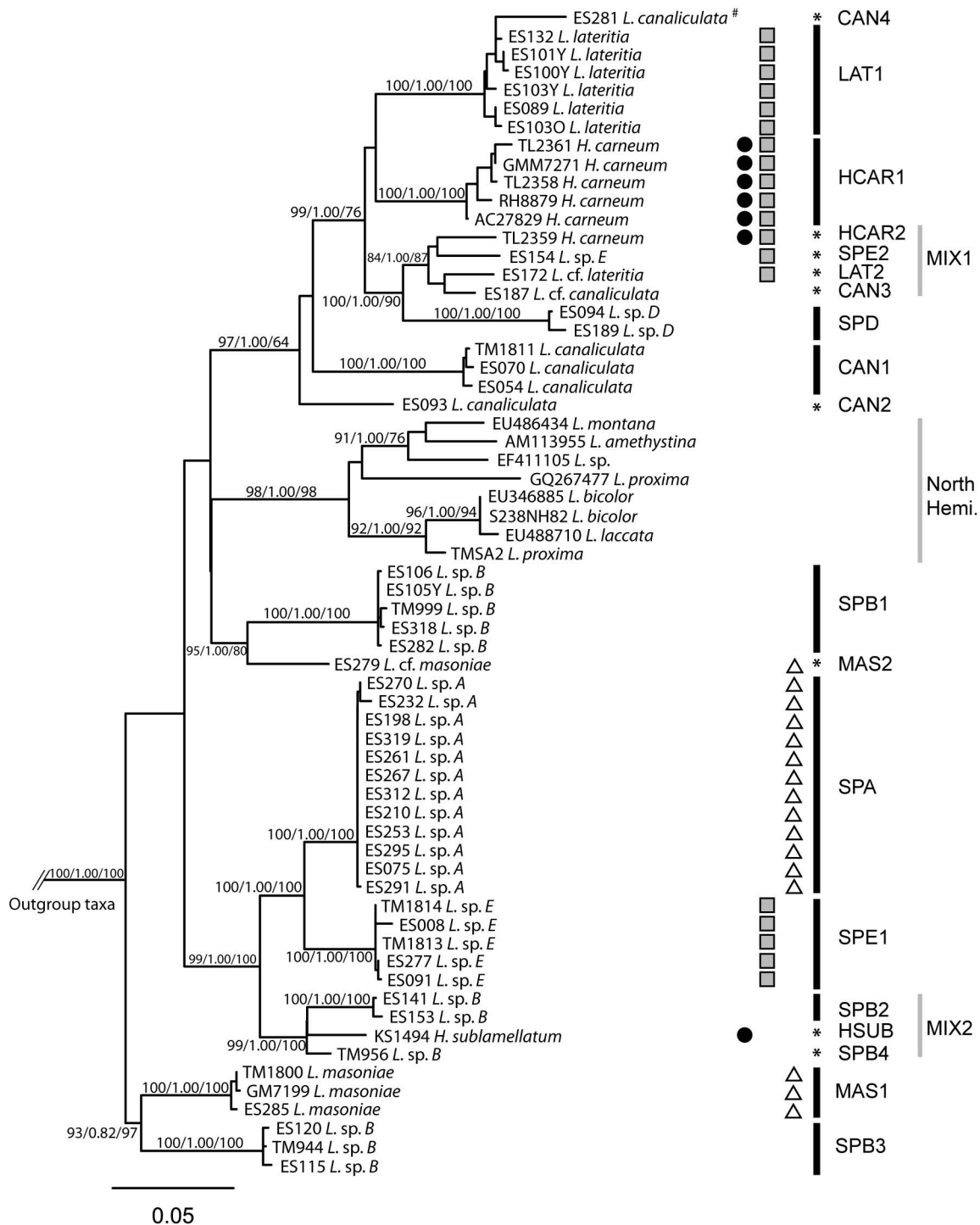


FIG. 1. Maximum likelihood tree of the concatenated internal transcribed spacer (ITS) of nuclear rDNA, RNA polymerase beta subunit (*rpb2*) and translation elongation factor 1 alpha (*tef1 α*) alignment. Values on branches are maximum likelihood/Bayesian inference/maximum parsimony, bootstrap support from 1000 replicates and posterior probabilities from 15 000 000 generations and only values above 75% on branches longer than 0.01 are shown. * = singletons, black circles = sequestrate collections, gray squares = two-spored collections, white triangles = collections under *Nothofagus cunninghamii*. Lineages are labeled with a three- or four-letter abbreviation for the relevant morphological species, followed by a number in cases where there are phylogenetic species within morphological species (as in SPB1, SPB2 etc.). Morphological species abbreviations are: *Hydnangium carneum* (HCAR); *H. sublamellatum* (HSUB); *Podohydangium australe* (PAUS); *Laccaria canaliculata* (CAN); *L. lateritia* (LAT); *L. masoniae* (MAS); *L. sp. A* (SPA); *L. sp. B* (SPB); *L. sp. C*

included in the patristic distance analysis because its position varied between trees. However in the *tef1 α* gene tree where ES281 was the sister collection to LAT1, these taxa were separated by a distance of >4%, which was much greater than the maximum intraspecies distance for *tef1 α* of 3.1%. The patristic distances between singleton collections and their nearest neighbor collection are included (SUPPLEMENTARY TABLE III).

Pairwise and patristic distances also were used to compare intra- and interspecies variation for the concatenated data and across the three gene regions. With pairwise distance, the concatenated analysis showed an overlap of 0.4% between the maximum intra- and minimum interspecific distances (FIG. 3) but with patristic distance had a clear barcoding gap between intraspecies variation (<2%) and interspecies variation (>3%). With pairwise distance both *rpb2* and *tef1 α* had small barcoding gaps of approximately 0.2%, whereas the ITS had an overlap of 1.4%. With patristic distance only *rpb2* had any gap (0.4%), whereas *tef1 α* had a small overlap (0.1%) and ITS had a large overlap (1.6%). The ITS had much less difference between species than the other gene regions based on both pairwise and patristic distance measures.

Values of 1, 2 and 3% dissimilarity of pairwise distances were used to cluster ITS sequences that represented 19 phylogenetic species. At 3% dissimilarity, clustering lumped most sequences together into a single taxonomic unit with only the phylogenetic species CAN1, SPD, HCARI and SPE2 (ES154) retained separately. The 2% cutoff also recovered these four phylogenetic species as well as SPA and lumped the remaining sequences into a further three taxonomic units. The 1% cutoff resulted in 18 taxonomic units and included two sequences that had been split from the phylogenetic species HCARI and three taxonomic units that contained two phylogenetic species each.

Species boundaries based on internal transcribed spacer gene tree with additional sequences.—The extended ITS gene tree included two morphological species of *Laccaria* from Australia that were not in the multigene phylogeny: *L. sp. C* (TM636) and *Podohydangium australe* (GQ981532). The position of the single *L. sp. C* collection was not supported in the ITS gene

tree although it associated more closely with the collections from SPB1 than any of the other species clades defined by the multigene phylogeny (SUPPLEMENTARY TABLE III). Another collection close to *L. sp. C* was a *L. laccata* sequence from GenBank (AJ699075); this collection probably had been misidentified because it was present under planted *Eucalyptus* (in India) and so is likely to be an Australian species that was introduced with the *Eucalyptus*, instead of *L. laccata*. *Podohydangium australe* was represented by a sequence from GenBank of a paratype collection that had moderate to high support (74% ML bootstrap, 1.00 posterior probability, 85% MP bootstrap) as a sister taxon to a *Hydnangium* sequence from New Zealand (FJ168595).

There was generally less support for the phylogenetic species in the extended ITS gene tree than in the tree based on the smaller ITS data set (TABLE III). There were also more inconsistencies between the three tree-building methods. Of the 10 clades identified in the multigene phylogeny and the constituent ITS gene tree, only seven were recovered in the expanded ITS gene tree with any support across all analyses (SPA, SPB2, SPB3, SPD, CAN1, LAT1, HCARI). In the extended ITS tree, the remaining three phylogenetic species (SPB1, MAS1, SPE1) all failed to resolve using at least one of the tree-building methods, and MAS1 was resolved only as paraphyletic with SPB3 (BI, ML) and not resolved at all with MP. The mixed species clade, MIX1, was polyphyletic in the expanded ITS tree under all tree-building methods. Six of the eight potential species identified in the multigene phylogeny from singletons could not be differentiated in the extended ITS gene tree, with patristic distances to the nearest neighbor of <2% (SUPPLEMENTARY TABLE III). Three putative species that were not in the multigene phylogeny but were differentiated as singletons in the extended ITS gene-tree were *P. australe* and two unidentified sequences from ectomycorrhizae in Australia (JF960795, JF960796). The latter sequences formed a moderate to highly supported pair (94% ML bootstrap, 0.99 posterior probability, 75% MP bootstrap) with both collections on the end of long branches.

In the extended ITS dataset, additional sequences that associated with clades from the multigene

←

(SPC); *L. sp. D* (SPD); and *L. sp. E* (SPE). The labels MIX1 and MIX2 correspond to clades of mixed species referred to in the text and North Hemi. = sequences from northern hemisphere species. In the parsimony tree LAT1 had 90% bootstrap support with collection ES281 as sister. Singleton collections that had been identified as close to (cf.) particular species also are designated by the letter abbreviation of the species.

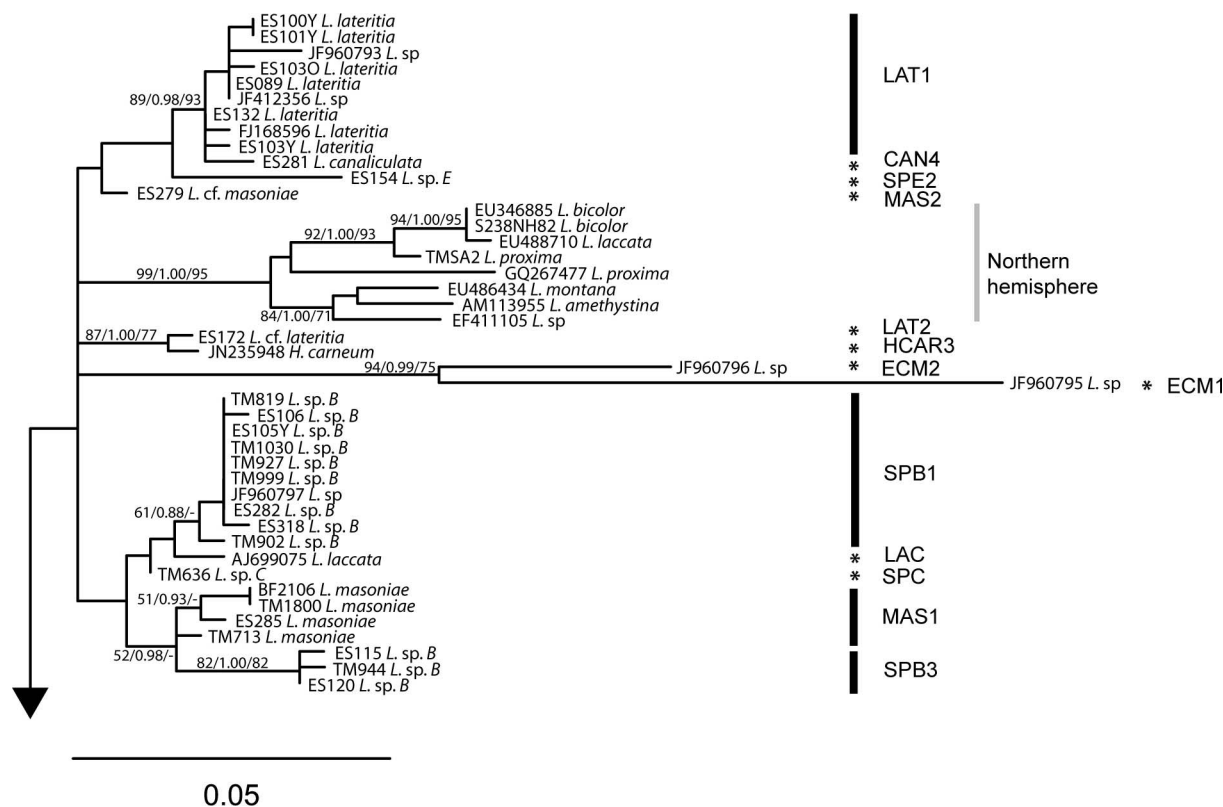


FIG. 2. Maximum likelihood tree of the internal transcribed spacer (ITS) of nuclear rDNA. Values on branches are maximum likelihood/Bayesian inference/maximum parsimony bootstrap support from 1000 replicates and posterior probabilities from 15 000 000 generations and only values above 75% on branches longer than 0.01 or that subtend phylogenetic species clades (FIG. 1) are shown. Lineage labels are as in FIG. 1 plus the following: *Laccaria laccata* (LAC), probably misidentified; *L. spp* from ectomycorrhizal root tip samples (ECM); and *L. glabripes*, *L. spp* and *H. sp.* from New Zealand (NZ).

phylogeny were 12 collections with SPA, five with SPB1, two with SPD, two with MAS1, four with LAT1, one with CAN1 and two with HCAR1; one collection (TM188) was also in the vicinity of the MIX2 clade. No additional collections were present in the SPB3 or SPE1 clades. Additional collections that had been assigned to morphological species in this study all matched the morphological species comprising the clades with which they associated. Two unidentified collections from GenBank were assigned to the clade LAT1 (JF412356, JF960793), one to the clade HCAR1 (JF960794) and one to SPD (FJ168606). Sequences from New Zealand did not form a monophyletic group and did not fall within any of the established clades of Australian species. Sequestrate collections also did not form a monophyletic group.

DISCUSSION

Species boundaries of Laccaria from southern Australia.— This paper describes a comprehensive analysis of 175 new DNA sequences from 77 collections of Hydngiaceae native to Australia. By using a multigene

phylogeny and the combination of genealogical concordance, sequence divergence and morphological uniformity within clades, 18 putative species of *Laccaria* (inclusive of *Hydnangium* and *Podohydngium*) from southern Australia were detected. Eight of these species were present only as singletons. When additional sequences from the expanded ITS gene tree are taken into account, a further three putative species from Australia were identified, bringing the total to 21 putative species from native habitats. Given that only 12 morphological species of Hydngiaceae are currently recognized from such habitats, this indicates a relatively high amount of cryptic speciation, especially considering that the original delimitation on morphology was based on multivariate analysis of an extensive set of macro- and micro-characters. The use of morphological characters in the original identification of collections was important to direct taxon sampling, rather than sequencing randomly from the extensive collections of *Laccaria* already in herbaria.

Molecular phylogenies can reveal two major patterns with relation to morphological species:

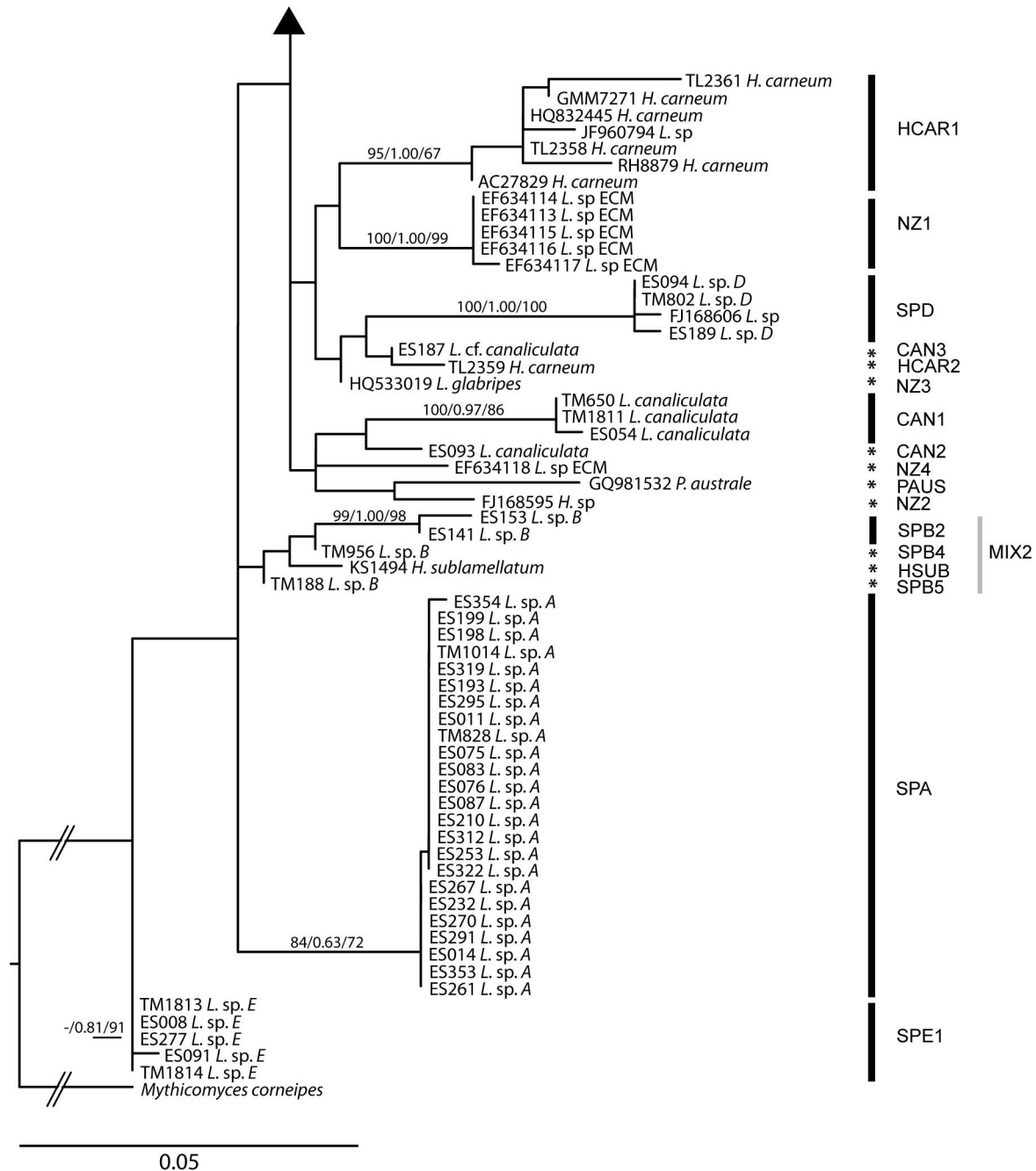


FIG. 2. Continued.

the combination of multiple species into a single morphologically variable species and the separation of species into undetected cryptic species (or pseudo-cryptic species if morphological or other characters are later found to support species distinctions). In this study it is noteworthy that there was no mixing in any of the phylogenetic species of different morphological species.

Reports of cryptic species have increased since use of molecular sequence data has become more

frequent (Bickford et al. 2007). In this study, generally low numbers of cryptic species are associated with the majority of the morphological species, which is in contrast to other studies where most of the cryptic species are found within a single morphological species or species complex as in *Hymenogaster* (Stielow et al. 2011), *Serpula himantoides* (Carlsen et al. 2011) and *Lactarius volemus* (Van de Putte et al. 2010). Detecting cryptic species can have an impact on conservation ratings and practices as for some

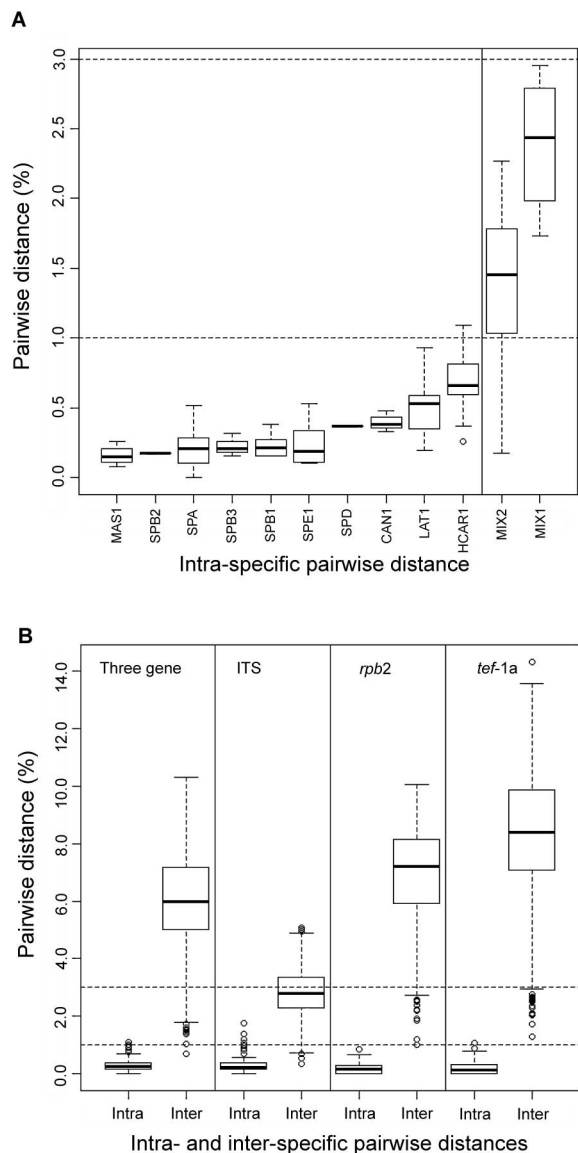


FIG. 3. Pairwise genetic distances between collections of Australian Hydnangiaceae. Standard boxplots with outliers indicated by circles. A. Intraspecific variation from the three-gene alignment of internal transcribed spacer (ITS) of nuclear rDNA; RNA polymerase beta subunit (*rpb2*); and translation elongation factor 1 alpha (*tef1 α*). Species were defined by clade boundaries on the multigene tree. Labels as in FIG. 1. B. Intra- and interspecific variation comparison across the three gene regions.

British *Hydnellum* and *Phellodon* species that are potential red list targets (Ainsworth et al. 2010). While *Laccaria* species look highly similar and are generally regarded as abundant, the presence of multiple cryptic lineages may reveal rare species that are in need of protection despite the perception of being common.

Cryptic species can be detected across barriers such as geography (e.g. within *Mycena pura*, Harder et al.

2010; within *Laccaria amethystina*, Vincenot et al. 2011) and there may well be a geographical component to some of the cryptic species detected in this study. The three phylogenetic species (LAT2, SPE2, SPB2) collected from the Australian Capital Territory, which was the most northern site included, were phylogenetically distinct from other collections assigned on morphology to *L. lateritia*, *L. sp. E* and *L. sp. B*. Two sister cryptic species of the Australian *Lactarius wirrabara* (Stubbe et al. 2011) were similarly divided into a Queensland and a Victoria/Tasmania lineage but were too under-sampled to confirm this pattern. While the collections for this study were predominately from Victoria and Tasmania, *Laccaria* species occur in all states of Australia and collections from the entire distribution will provide a greater understanding of potential phylogeographic patterns as well as of the total diversity in Australia. Large scale sequencing of collections currently housed in herbaria is one way to substantially increase taxon sampling (e.g. *Cortinarius*, Harrower et al. 2011).

The additional sequences in the expanded ITS gene tree produced eight lineages that were not detected by the more limited multigene phylogeny. Only two of these were from described morphological species, *L. sp. C* and *P. australe*. The sequences on the two longest branches in the tree (JF960796, JF960795) also represent unique lineages of Australian *Laccaria*. The other new lineages were from the GenBank sequences of New Zealand collections, only one of which, HQ533019 *L. glabripes*, was identified to species. The morphological character of spore number per basidium (two or four) was not mixed in any of the phylogenetic species clades as accepted herein. Although May (1991) suggested there may be sister-species pairs between two- and four-spored taxa that otherwise were very similar (*L. sp. E* and *L. sp. A*; *L. lateritia* and *L. canaliculata*; *L. sp. C* and *L. sp. B*), this does not appear to be the case. Most of the taxa in these species pairs contain multiple cryptic species that are interspersed throughout the tree. However, *L. sp. A* and *L. sp. E* (SPE1) are strongly supported as sister species, *L. sp. C* associates closely with one of the *L. sp. B* clades (SPB1) and one of the *L. canaliculata* phylogenetic species (CAN4, ES281) is sister to one of the *L. lateritia* clades (LAT1).

The multigene phylogeny recovered a clade of northern hemisphere species that nested within the Australian clades or formed a polytomy with them if branches without support were collapsed. This is congruent with other research that included *Laccaria* species from both the northern and southern hemispheres (Ryberg and Matheny 2012) and found a disjunction between the two. The split between taxa from the north and taxa from Australia and New

Zealand was approximately 40 000 000 y ago (Ryberg and Matheny 2012) and it seems there has been little if any movement of species between these two regions since.

Barcode for species of Australian Laccaria.—While there is no single cutoff value that is suitable for distinguishing species clades across all fungi (Nilsson et al. 2008) a standard value of 3% sequence dissimilarity of ITS is often used to define molecular operational taxonomic units (MOTUs; Hibbett et al. 2011), which sometimes are designated as species (e.g. Ryberg et al. 2008, Brock et al. 2009, Hughes et al. 2009, Harrower et al. 2011). In *Laccaria* a cutoff value of 2% has been used to define taxonomic units of environmental ITS sequences (Tedersoo et al. 2009). While the ITS was determined herein to be unable to distinguish between some species, a cutoff value 1–2% would better approximate species of Australian *Laccaria* than 2% or 3%, although greater sampling of species is required to refine this value. A large barcoding gap was found only by using patristic distance on the multigene phylogeny, and in this case a 2% cutoff was sufficient to contain the intraspecific variation for the majority of phylogenetic species. If any one of the three gene regions tested here were to be used alone to distinguish species, the *rpb2* should be considered because it performed best with both pairwise and patristic distances and is easy to align due to a lack of indels. Unfortunately we were unable to amplify the *rpb2* region from older herbarium material and it lacked a highly conserved motif that would let it be split into smaller regions that may have been amplified individually, as is the case with the ITS (ITS1, 5.8S, ITS2). The ITS tree was the least informative of the three gene regions with little variation between some species. While the ITS is not the ideal barcoding gene for Australian *Laccaria*, it was still useful in assigning likely species designations for some unidentified GenBank sequences.

Synonymy of Hydnangium and Podohydangium with Laccaria.—In this study *Hydnangium* was represented by 11 specimens from Australia and the monotypic *Podohydangium* by the sole sequence in GenBank (a paratype collection from the Otway Ranges in Victoria). Our finding that these genera are synonymous with *Laccaria* is consistent with Mueller and Pine (1994), who used partial ITS sequences to place both *Hydnangium* and *Podohydangium* within *Laccaria*. While *Hydnangium* is clearly a synonym of *Laccaria*, the latter is actually the more recent name. Due to the more numerous species described in *Laccaria* and its use as a model ectomycorrhizal fungus, it will be preferable to conserve the name *Laccaria* over *Hydnangium*.

The collections of *Hydnangium* and *P. australe* did not form a monophyletic group within the ITS gene tree, which suggests the sequestrate basidiocarp form has evolved multiple times in Australian *Laccaria*. The ITS2 gene tree by Mueller and Pine (1994) had insufficient resolution and too few Australian *Laccaria* collections to be able to detect the pattern of multiple origins. Non-monophyly is concordant with other agaricoid genera within which sequestrate forms have evolved multiple times (e.g. *Cortinarius*, Peintner et al. 2001; *Russula*, Lebel and Tonkin 2007; *Amanita*, Justo et al. 2010). Reversals from the sequestrate basidiocarp form to the agaricoid form are considered highly unlikely to occur due to the difficulty of re-evolving the complex mechanism for forcible-spore discharge (Thiers 1984), and so far there is no evidence for this phenomenon.

Association of Laccaria with Nothofagus cunninghamii.—*Laccaria* sp. A and *L. masoniae* have been collected only from cool temperate rainforest sites in association with *Nothofagus cunninghamii*, which is the dominant ectomycorrhizal tree in these habitats. These two species occur in three separate positions in the phylogeny, and in each case the phylogenetic species are sister to clades containing species not recorded from *Nothofagus* forest, such as *L.* sp. B. This pattern is consistent with multiple gains or losses of the association with *N. cunninghamii*. Multiple host switches between Nothofagaceae and Myrtaceae have been reported in the Hysterangiales, with more switches to than from Nothofagaceae (Hosaka et al. 2008).

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LITERATURE CITED

- Ainsworth AM, Parfitt D, Rogers HJ, Boddy L. 2010. Cryptic taxa within European species of *Hydnellum* and *Phellodon* revealed by combined molecular and morphological analysis. *Fungal Ecol* 3:65–80, doi:10.1016/j.funeco.2009.07.001
- Beaton G, Pegler DN, Young TWK. 1984. Gasteroid Basidiomycota of Victoria state, Australia I. Hydnangiaceae. *Kew Bull* 39:499–508, doi:10.2307/4108593

- Bickford D, Lohman DJ, Sodhi NS, Ng PKL, Meier R, Winker K, Ingram KK, Das I. 2007. Cryptic species as a window on diversity and conservation. *Trends Ecol Evol* 22:148–155, doi:10.1016/j.tree.2006.11.004
- Bougher NL, Tommerup IC, Malajczuk N. 1993. Broad variation in developmental and mature basidiome morphology of the ectomycorrhizal fungus *Hydnangium sublamellatum* sp. nov. bridges morphologically based generic concepts of *Hydnangium*, *Podohydangium* and *Laccaria*. *Mycol Res* 97:613–619, doi:10.1016/S0953-7562(09)81186-6
- Brock PM, Döring H, Bidartondo MI. 2009. How to know unknown fungi: the role of a herbarium. *New Phytol* 181:719–724, doi:10.1111/j.1469-8137.2008.02703.x
- Carlsen T, Engh IB, Decock C, Rajchenberg M, Kausarud H. 2011. Multiple cryptic species with divergent substrate affinities in the *Serpula himantioides* species complex. *Fungal Biol* 115:54–61, doi:10.1016/j.funbio.2010.10.004
- The Council of Heads of Australasian Herbaria. (1999–). Australia's Virtual Herbarium www.chah.gov.au/avh
- Fourment M, Gibbs MJ. 2006. Patristic: a program for calculating patristic distances and graphically comparing the components of genetic change. *BMC Evol Biol* 6:1, doi:10.1186/1471-2148-6-1
- Harder CB, Læssøe T, Kjoller R, Frøslev TG. 2010. A comparison between ITS phylogenetic relationships and morphological species recognition within *Mycena* sect. *Calodontes* in northern Europe. *Mycol Prog* 9:395–405, doi:10.1007/s11557-009-0648-7
- Harrower E, Ammirati JF, Cappuccino AA, Ceska O, Kranabetter JM, Kroeger P, Lim S, Taylor T, Berbee ML. 2011. *Cortinarius* species diversity in British Columbia and molecular phylogenetic comparison with European specimen sequences. *Botany* 89:799–810, doi:10.1139/b11-065
- Hawksworth DL. 2006. Pandora's mycological box: molecular sequences vs. morphology in understanding fungal relationships and biodiversity. *Rev Iberoam Micol* 23:127–133, doi:10.1016/S1130-1406(06)70031-6
- Hibbett DS, Ohman A, Glotzer D, Nuhn M, Kirk P, Nilsson RH. 2011. Progress in molecular and morphological taxon discovery in Fungi and options for formal classification of environmental sequences. *Fungal Biol Rev* 25:38–47, doi:10.1016/j.fbr.2011.01.001
- Hosaka K, Castellano MA, Spatafora JW. 2008. Biogeography of Hysterangiales (Phallomycetidae, Basidiomycota). *Mycol Res* 112:448–462, doi:10.1016/j.mycres.2007.06.004
- Hughes KW, Petersen RH, Lickey EB. 2009. Using heterozygosity to estimate a percentage DNA sequence similarity for environmental species delimitation across basidiomycete fungi. *New Phytol* 189:795–798, doi:10.1111/j.1469-8137.2009.02802.x
- Jombart T, Balloux F, Dray S. 2010. adephylo: new tools for investigating the phylogenetic signal in biological traits. *Bioinformatics* 26:1907–1909, doi:10.1093/bioinformatics/btq292
- Justo A, Morgenstern I, Hallen-Adams HE, Hibbett DS. 2010. Convergent evolution of sequestrate forms in *Amanita* under Mediterranean climate conditions. *Mycologia* 102:675–688, doi:10.3852/09-191
- Kühner R. 1980. Les Hyménomycètes agaricoïdes. *Bull Mens Soc Linn Lyon* 49:1–1027.
- Lebel T, Tonkin JE. 2007. Australasian species of *Macowanites* are sequestrate species of *Russula* (Russulaceae, Basidiomycota). *Aust Syst Bot* 20:355–381, doi:10.1071/SB07007
- Lefébure T, Douady CJ, Gouy M, Gibert J. 2006. Relationship between morphological taxonomy and molecular divergence within Crustacea: proposal of a molecular threshold to help species delimitation. *Mol Phylogenet Evol* 40:435–447, doi:10.1016/j.ympev.2006.03.014
- Maddison WP. 1997. Gene trees in species trees. *Syst Biol* 46:523–536, doi:10.1093/sysbio/46.3.523
- Martin F, Aerts A, Ahren D, Brun A, Danchin EGJ, Duchaussoy F, Gibon J, Kohler A, Lindquist E, Pereda V, Salamov A, Shapiro HJ, Wuyts J, Blaudez D, Buee M, Brokstein P, Canback B, Cohen D, Courty PE, Coutinho PM, Delaruelle C, Detter JC, Deveau A, DiFazio S, Duplessis S, Fraissinet-Tachet L, Lucic E, Frey-Klett P, Fourrey C, Feussner I, Gay G, Grimwood J, Hoegger PJ, Jain P, Kilaru S, Labbe J, Lin YC, Legue V, Le Tacon F, Marmeisse R, Melayah D, Montanini B, Muratet M, Nehls U, Niculita-Hirzel H, Oudot-Le Secq MP, Peter M, Quesneville H, Rajashekar B, Reich M, Rouhier N, Schmutz J, Yin T, Chalot M, Henrissat B, Kues U, Lucas S, van de Peer Y, Podila GK, Polle A, Pukkila PJ, Richardson PM, Rouze P, Sanders IR, Stajich JE, Tunlid A, Tuskan G, Grigoriev IV. 2008. The genome of *Laccaria bicolor* provides insights into mycorrhizal symbiosis. *Nature* 452:88–92, doi:10.1038/nature06556
- , Díez J, Dell B, Delaruelle C. 2002. Phylogeography of the ectomycorrhizal *Pisolithus* species as inferred from nuclear ribosomal DNA ITS sequences. *New Phytol* 153:345–357, doi:10.1046/j.0028-646X.2001.00313.x
- Matheny PB. 2005. Improving phylogenetic inference of mushrooms with RPB1 and RPB2 nucleotide sequences (*Inocybe*, Agaricales). *Mol Phylogenet Evol* 35:1–20, doi:10.1016/j.ympev.2004.11.014
- , Curtis JM, Hofstetter V, Aime MC, Moncalvo J-M, Ge ZW, Yang ZL, Slot JC, Ammirati JF, Baroni TJ, Bougher NL, Hughes KW, Lodge DJ, Kerrigan RW, Seidl MT, Aanen DK, DeNitis M, Daniele GM, Desjardin DE, Kropp BR, Norvell LL, Parker A, Vellinga EC, Vilgalys R, Hibbett DS. 2006. Major clades of Agaricales: a multilocus phylogenetic overview. *Mycologia* 98:982–995, doi:10.3852/mycologia.98.6.982
- May TW. 1991. A taxonomic study of the Australian species of *Laccaria* [doctoral dissertation]. Monash Univ. Press. 339 p.
- . *Laccaria*. In: Grgurinovic CA. Larger fungi of South Australia. The botanic gardens of Adelaide and state herbarium and the flora and fauna. Adelaide: South Australia Handbooks Committee, 296–310.
- Mueller GM. 1992. Systematics of *Laccaria* (Agaricales) in the continental United States and Canada, with

- discussions on extralimital taxa and descriptions of extant types. *Fieldiana Bot* 30:1–158.
- , Hofsten AV, Axén A, Strack BA. 1993. Basidiospore wall ultrastructure of the false-truffle *Hydnangium* and its phylogenetic significance. *Mycologia* 85:890–893, doi:10.2307/3760670
- , Pine EM. 1994. DNA data provide evidence on the evolutionary relationships between mushrooms and false truffles. *McIlvainea* 11:61–74.
- Nilsson RH, Kristiansson E, Ryberg M, Hallenberg N. 2008. Intraspecific ITS variability in the kingdom Fungi as expressed in the international sequence databases and its implications for molecular species identification. *Evol Bioinform* 4:193–201.
- Nylander JAA, Wilgenbusch JC, Warren DL, Swofford DL. 2008. AWTY (are we there yet?): a system for graphical exploration of MCMC convergence in Bayesian phylogenetics. *Bioinformatics* 24:581–583, doi:10.1093/bioinformatics/btm388
- Osmundson TW, Cripps CL, Mueller GM. 2005. Morphological and molecular systematics of Rocky Mountain alpine *Laccaria*. *Mycologia* 97:949–972, doi:10.3852/mycologia.97.5.949
- Peintner U, Bougher NL, Castellano MA, Moncalvo JM, Moser MM, Trappe JM, Vilgalys R. 2001. Multiple origins of sequestrate fungi related to *Cortinarius* (Cortinariaceae). *Am J Bot* 88:2168–2179, doi:10.2307/3558378
- Van de Putte K, Nuytinck J, Das K, Verbeken A. 2012. Exposing hidden diversity by concordant genealogies and morphology—a study of the *Lactifluus volemus* (Russulales) species complex in Sikkim Himalaya (India). *Fungal Divers* 55:171–194, doi:10.1007/s13225-012-0162-0
- , ———, Stubbe D, Le HT, Verbeken A. 2010. *Lactarius volemus* sensu lato (Russulales) from northern Thailand: morphological and phylogenetic species concepts explored. *Fungal Divers* 45:99–130, doi:10.1007/s13225-010-0070-0
- Rehner SA, Buckley E. 2005. A *Beauveria* phylogeny inferred from nuclear ITS and EF1- α sequences: evidence for cryptic diversification and links to *Cordyceps* teleomorphs. *Mycologia* 97:84–98, doi:10.3852/mycologia.97.1.84
- Ronquist F, Huelsenbeck JP. 2003. MrBayes 3: Bayesian phylogenetic inference under mixed models. *Bioinformatics* 19:1572–1574, doi:10.1093/bioinformatics/btg180
- Rozen R, Skaletsky HJ. 2000. Primer3 on the WWW for general users and for biologist programmers. In: Krawetz S, Misener S, eds. *Bioinformatics methods and protocols: methods in molecular biology*. Totowa, New Jersey: Humana Press. p 365–386.
- Ryberg M, Matheny PB. 2012. Asynchronous origins of ectomycorrhizal clades of Agaricales. *P Roy Soc B-Biol Sci* 279:2003–2011, doi:10.1098/rspb.2011.2428
- , Nilsson RH, Kristiansson E, Töpel M, Jacobsson S, Larsson E. 2008. Mining metadata from unidentified ITS sequences in GenBank: a case study in *Inocybe* (Basidiomycota). *BMC Evol Biol* 8:50, doi:10.1186/1471-2148-8-50
- Schloss PD, Westcott SL, Ryabin T, Hall JR, Hartmann M, Hollisters EB, Lesniewski RA, Oakley BB, Parks DH, Robinson CJ, Sahl JW, Stres B, Thallinger GG, van Horn DJ, Weber CF. 2009. Introducing mothur: open-source, platform-independent, community-supported software for describing and comparing microbial communities. *Appl Environ Microbiol* 75:7537–7541, doi:10.1128/AEM.01541-09
- Singer R. 1986. The Agaricales in modern taxonomy. 4th ed. Koenigstein, Germany: Koeltz Scientific Books. 981 p.
- Stamatakis A, Hoover P, Rougemont J. 2008. A rapid bootstrap algorithm for the RAxML web servers. *Syst Biol* 57:758–771, doi:10.1080/10635150802429642
- Stielow B, Bratek Z, Orczán AKI, Rudnoy S, Hensel G, Hoffmann P, Klenk H-P, Göker M. 2011. Species delimitation in taxonomically difficult fungi: the case of *Hymenogaster*. *PloS ONE* 6:e15614, doi:10.1371/journal.pone.0015614
- Stubbe D, Le HT, Wang X-H, Nuytinck J, Van de Putte K, Verbeken A. 2011. The Australasian species of *Lactarius* subgenus *Gerardii* (Russulales). *Fungal Divers* 52:141–167, doi:10.1007/s13225-011-0111-3
- Swofford DL. 1998. PAUP* 4: phylogenetic analysis using parsimony (*and other methods). Sunderland, Massachusetts: Sinauer Associates.
- Tamura K, Dudley J, Nei M, Kumar S. 2007. MEGA4: molecular evolutionary genetics analysis (MEGA) software. *Mol Biol Evol* 24:1596–1599, doi:10.1093/molbev/msm092
- Taylor JW, Jacobson DJ, Kroken S, Kasuga T, Geiser DM, Hibbett DS, Fisher MC. 2000. Phylogenetic species recognition and species concepts in fungi. *Fungal Genet Biol* 31:21–32, doi:10.1006/fgbi.2000.1228
- Tedersoo L, Gates G, Dunk CW, Lebel T, May TW, Kõljalg U, Jairus T. 2009. Establishment of ectomycorrhizal fungal community on isolated *Nothofagus cunninghamii* seedlings regenerating on dead wood in Australian wet temperate forests: Does fruit-body type matter? *Mycorrhiza* 19:403–416, doi:10.1007/s00572-009-0244-3
- Thiers HD. 1984. The secotoid syndrome. *Mycologia* 76:1–8, doi:10.2307/3792830
- Thompson JD, Higgins DG, Gibson TJ. 1994. Clustal W: improving the sensitivity of progressive multiple sequence alignment through sequence weighting, position-specific gap penalties and weight matrix choice. *Nucleic Acids Res* 22:4673–4680, doi:10.1093/nar/22.22.4673
- Vincenot L, Nara K, Sthultz C, Labbé J, Dubois M-P, Tedersoo L, Martin F, Selosse M-A. 2011. Extensive gene flow over Europe and possible speciation over Eurasia in the ectomycorrhizal basidiomycete *Laccaria amethystina* complex. *Mol Ecol* 21:281–299, doi:10.1111/j.1365-294X.2011.05392.x
- White TJ, Bruns T, Lee S, Taylor J. 1990. Amplification and direct sequencing of fungal ribosomal RNA genes for phylogenetics. In: Innis MA, Gelfand DH, Sninsky JJ, White TJ, eds. *PCR protocols: a guide to methods and applications*. New York: Academic Press. p 315–322.